

Estimated Annual Agricultural Pesticide Use Franklin County, Washington 2014-2018

24 July 2026

Summary

Franklin County ranks **#3 of Washington's 39 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of WA's five highest pesticide-use counties. Franklin County, also part of the Tri-Cities area, is irrigated Columbia Basin farmland producing potatoes, wheat, alfalfa, and seed crops.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Franklin County, Washington** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Franklin County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	DTC	2040410	1989474	1575444
2	DICHLOROPROPENE	OC	1309165	1333244	1230414
3	METAM POTASSIUM	DTC	535658	346441	349962
4	CHLOROPICRIN	OC	251181	104628	69726
5	PETROLEUM OIL	OTH	234966	295702	281578
6	BACILLUS AMYLOLIQUEFACIEN	MIC	134949	92658	65722
7	SULFURIC ACID	INO	130767	282544	296825
8	SULFUR	INO	113966	89341	87078
9	CALCIUM POLYSULFIDE	INO	109971	58376	52081
10	GLYPHOSATE	OTH	47902	51402	50415
11	EPTC	TC	36515	31098	33234
12	MANCOZEB	DTC	35588	44621	54601
13	COPPER	INO	24253	15833	12841
14	CHLOROTHALONIL	OTH	24188	24370	24321
15	PARAQUAT	DIP	20229	14209	12433
16	PENDIMETHALIN	OTH	16362	14793	13419
17	OXAMYL	CB	15230	6080	11094
18	METRIBUZIN	TRZ	14046	15583	14343
19	METOLACHLOR & METOLACHLOR-S	OTH	11870	9741	9123
20	METOLACHLOR-S	OTH	11870	9741	9046
21	2,4-D	CPX	10476	11616	13517
22	DIMETHENAMID & DIMETHENAMID-P	OTH	10050	6984	5621
23	DIMETHENAMID-P	OTH	10050	6984	5621
24	BURKHOLDERIA SPP	MIC	9485	8303	6259
25	COPPER HYDROXIDE	INO	8211	8318	7231
26	LINURON	OTH	6962	3667	2437
27	DIMETHOATE	OP	6766	5508	5384
28	COPPER SULFATE	INO	6589	5333	4903
29	BACILLUS THURINGIENSIS	MIC	6544	3070	2452
30	DIURON	OTH	6459	4056	4009
31	DIQUAT	DIP	6061	4261	4291
32	GLUFOSINATE	OTH	5919	2798	2628
33	COPPER OXYCHLORIDE	INO	5188	3326	2325
34	MALEIC HYDRAZIDE	OTH	5016	5610	5904
35	CHLORPYRIFOS	OP	4520	7893	7393
36	THIOPHANATE-METHYL	OTH	4377	5131	7110
37	PENTHIOPYRAD	OTH	4185	2698	1940
38	ATRAZINE	TRZ	4184	5052	4967
39	FLUAZINAM	OTH	4155	3285	3303
40	BENTAZONE	OTH	4151	4415	5134
41	PHOSPHORIC ACID	INO	3887	3369	8713
42	METHOMYL	CB	3478	6382	4933
43	MEFENOXAM	OTH	3378	2956	2584
44	POTASSIUM BICARBONATE	INO	3028	3075	2400
45	BROMOXYNIL	OTH	2881	4114	5518
46	FLUTOLANIL	OTH	2715	1233	1671

Table 2: Top 80 EPest-high Estimates, Franklin County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	PROPICONAZOLE	OTH	2292	2652	1982
48	KAOLIN CLAY	INO	2274	38899	26050
49	HEXAZINONE	TRZ	2204	3168	3054
50	SPIROMESIFEN	OTH	2107	2003	2010
51	MCPA	CPX	2068	2878	3277
52	PROPARGITE	OTH	1864	1872	2657
53	CYMOXANIL	OTH	1812	1707	1241
54	BOSCALID	OTH	1810	2843	3683
55	CARBARYL	CB	1754	2216	3554
56	SPIROTETRAMAT	OTH	1496	1253	1283
57	PYRACLOSTROBIN	OTH	1468	2152	2183
58	BUPROFEZIN	OTH	1344	892	722
59	PYMETROZINE	OTH	1331	800	880
60	2,4-DB	CPX	1312	1209	965
61	AZOXYSTROBIN	OTH	1242	2124	1802
62	ETHALFLURALIN	OTH	1225	1689	2438
63	AUREOBASIDIUM PULLULANS	MIC	1189	868	524
64	TRIFLUMIZOLE	OTH	1188	1333	1298
65	BIFENTHRIN	PYR	1187	1059	819
66	MCPB	CPX	1165	600	403
67	NORFLURAZON	OTH	1099	778	1327
68	TRIFLURALIN	OTH	1059	1215	1176
69	DCPA	OTH	1037	4158	4328
70	IMIDACLOPRID	OTH	1030	985	1433
71	CLETHODIM	OTH	961	690	747
72	DIMETHOMORPH	OTH	944	583	455
73	SIMAZINE	TRZ	932	345	365
74	MALATHION	OP	854	949	1886
75	THIAMETHOXAM	OTH	818	1066	1067
76	FLUMIOXAZIN	OTH	807	670	452
77	ZOXAMIDE	OTH	799	400	718
78	CHLORANTRANILIPROLE	OTH	757	662	687
79	PROHEXADIONE	OTH	736	358	472
80	ETHOFUMESATE	OTH	731	602	461

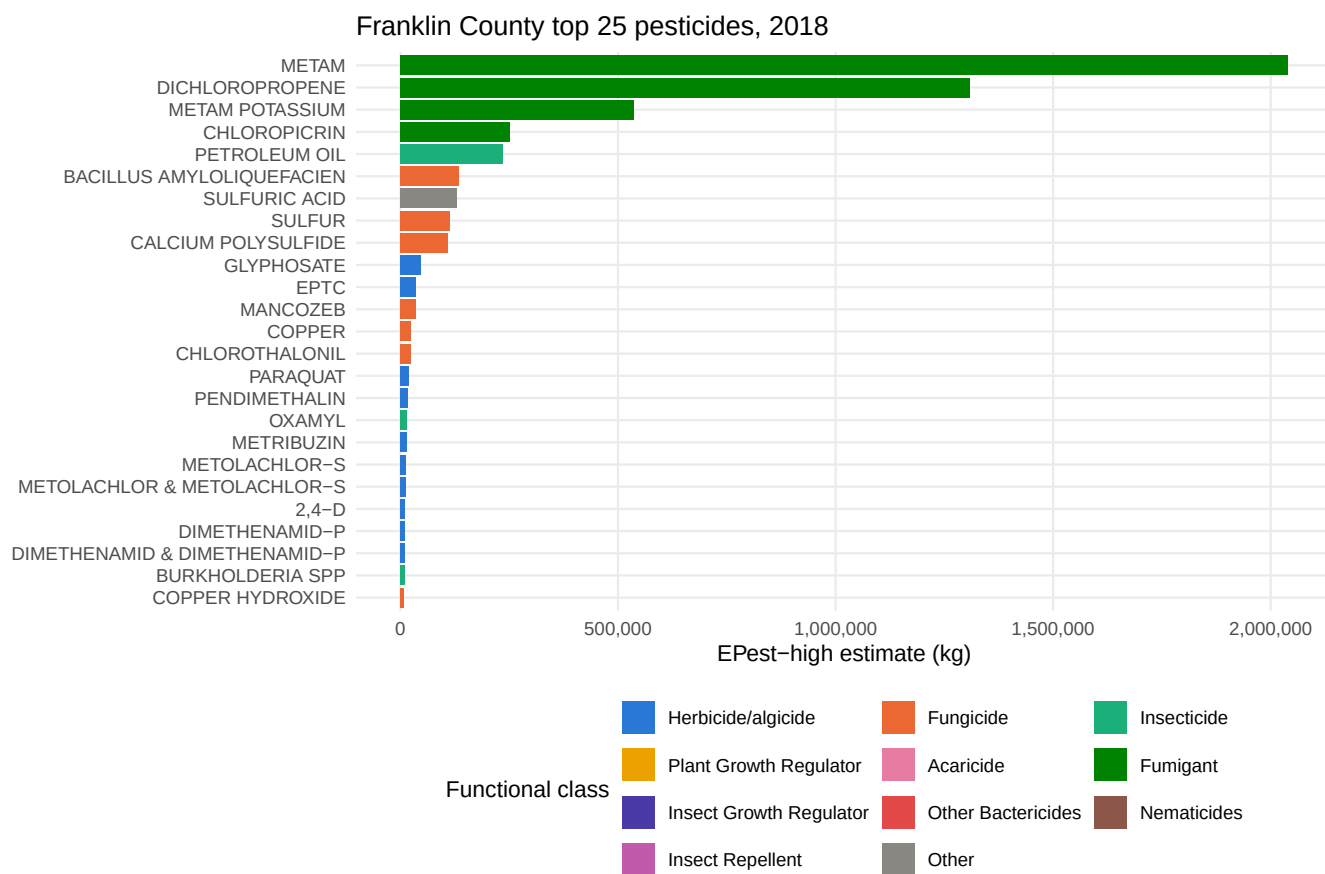


Figure 1: Top 25 compounds in Franklin County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Franklin County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	2040410	2040410
2	DICHLOROPROPENE	1309165	1309165
3	METAM POTASSIUM	535658	535658
4	CHLOROPICRIN	0	251181
5	PETROLEUM OIL	233373	234966
6	BACILLUS AMYLOLIQUEFACIEN	134949	134949
7	SULFURIC ACID	0	130767
8	SULFUR	113820	113966
9	CALCIUM POLYSULFIDE	109964	109971
10	GLYPHOSATE	47216	47902
11	EPTC	36515	36515
12	MANCOZEB	35237	35588
13	COPPER	24153	24253
14	CHLOROTHALONIL	24188	24188
15	PARQUAT	20052	20229
16	PENDIMETHALIN	14568	16362
17	OXAMYL	15230	15230
18	METRIBUZIN	12804	14046
19	METOLACHLOR & METOLACHLOR-S	11870	11870
20	METOLACHLOR-S	11870	11870
21	2,4-D	9949	10476
22	DIMETHENAMID & DIMETHENAMID-P	7834	10050
23	DIMETHENAMID-P	7834	10050
24	BURKHOLDERIA SPP	8628	9485
25	COPPER HYDROXIDE	8092	8211
26	LINURON	6918	6962
27	DIMETHOATE	6766	6766
28	COPPER SULFATE	6394	6589
29	BACILLUS THURINGIENSIS	6542	6544
30	DIURON	6436	6459
31	DIQUAT	6061	6061
32	GLUFOSINATE	5919	5919
33	COPPER OXYCHLORIDE	5188	5188
34	MALEIC HYDRAZIDE	5016	5016
35	CHLORPYRIFOS	4039	4520
36	THIOPHANATE-METHYL	3947	4377
37	PENTHIOPYRAD	3969	4185
38	ATRAZINE	4184	4184
39	FLUAZINAM	4155	4155
40	BENTAZONE	4151	4151
41	PHOSPHORIC ACID	3887	3887
42	METHOMYL	3478	3478
43	MEFENOXAM	3296	3378
44	POTASSIUM BICARBONATE	90	3028
45	BROMOXYNIL	2881	2881
46	FLUTOLANIL	2715	2715
47	PROPICONAZOLE	2292	2292
48	KAOLIN CLAY	1800	2274
49	HEXAZINONE	2204	2204
50	SPIROMESIFEN	2107	2107

Table 4: Top 50 Franklin County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	1989474	1989474
2	DICHLOROPROPENE	1333153	1333244
3	METAM POTASSIUM	346441	346441
4	PETROLEUM OIL	287727	295702
5	SULFURIC ACID	0	282544
6	CHLOROPICRIN	19633	104628
7	BACILLUS AMYLOLIQUEFACIEN	92658	92658
8	SULFUR	87508	89341
9	CALCIUM POLYSULFIDE	58374	58376
10	GLYPHOSATE	48524	51402
11	MANCOZEB	44504	44621
12	KAOLIN CLAY	4327	38899
13	EPTC	31098	31098
14	CHLOROTHALONIL	24361	24370
15	COPPER	15800	15833
16	METRIBUZIN	14999	15583
17	PENDIMETHALIN	12556	14793
18	PARAQUAT	13343	14209
19	2,4-D	10803	11616
20	METOLACHLOR & METOLACHLOR-S	6016	9741
21	METOLACHLOR-S	6016	9741
22	COPPER HYDROXIDE	8048	8318
23	BURKHOLDERIA SPP	7884	8303
24	CHLORPYRIFOS	5516	7893
25	ETHOPROPHOS	7722	7722
26	DIMETHENAMID & DIMETHENAMID-P	6137	6984
27	DIMETHENAMID-P	6137	6984
28	METHOMYL	6382	6382
29	OXAMYL	5813	6080
30	ZIRAM	0	5709
31	MALEIC HYDRAZIDE	5610	5610
32	DIMETHOATE	5424	5508
33	COPPER SULFATE	4650	5333
34	THIOPHANATE-METHYL	4698	5131
35	CUPROUS OXIDE	4963	5070
36	ATRAZINE	5052	5052
37	BENTAZONE	4415	4415
38	DIQUAT	4261	4261
39	DCPA	4158	4158
40	BROMOXYNIL	3424	4114
41	DIURON	3213	4056
42	LINURON	3389	3667
43	PHOSPHORIC ACID	2176	3369
44	COPPER OXYCHLORIDE	3326	3326
45	FLUAZINAM	3285	3285
46	HEXAZINONE	3168	3168
47	POTASSIUM BICARBONATE	822	3075
48	BACILLUS THURINGIENSIS	3069	3070
49	MEFENOXAM	2929	2956
50	MCPA	2180	2878

Table 5: Top 50 Franklin County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	1575444	1575444
2	DICHLOROPROPENE	1229827	1230414
3	METAM POTASSIUM	346052	349962
4	SULFURIC ACID	0	296825
5	PETROLEUM OIL	276793	281578
6	SULFUR	82875	87078
7	CHLOROPICRIN	16068	69726
8	BACILLUS AMYLOLIQUEFACIEN	65722	65722
9	MANCOZEB	54443	54601
10	CALCIUM POLYSULFIDE	50592	52081
11	GLYPHOSATE	47705	50415
12	EPTC	32620	33234
13	KAOLIN CLAY	5307	26050
14	CHLOROTHALONIL	24311	24321
15	METRIBUZIN	13798	14343
16	2,4-D	10240	13517
17	PENDIMETHALIN	11650	13419
18	COPPER	12464	12841
19	PARAQUAT	11579	12433
20	OXAMYL	10934	11094
21	METOLACHLOR & METOLACHLOR-S	6781	9123
22	METOLACHLOR-S	6739	9046
23	PHOSPHORIC ACID	1347	8713
24	CHLORPYRIFOS	5460	7393
25	COPPER HYDROXIDE	6979	7231
26	THIOPHANATE-METHYL	6603	7110
27	BURKHOLDERIA SPP	5913	6259
28	MALEIC HYDRAZIDE	5904	5904
29	DIMETHENAMID & DIMETHENAMID-P	4931	5621
30	DIMETHENAMID-P	4931	5621
31	BROMOXYNIL	3447	5518
32	CUPROUS OXIDE	3820	5393
33	DIMETHOATE	5203	5384
34	BENTAZONE	5134	5134
35	ATRAZINE	4967	4967
36	METHOMYL	4694	4933
37	COPPER SULFATE	3722	4903
38	ETHOPROPHOS	4482	4482
39	DCPA	2891	4328
40	DIQUAT	4291	4291
41	PINOLENE	4029	4029
42	DIURON	3253	4009
43	BOSCALID	3617	3683
44	METIRAM	2789	3649
45	CARBARYL	3502	3554
46	ZIRAM	591	3446
47	FLUAZINAM	3303	3303
48	MCPA	2856	3277
49	HEXAZINONE	3054	3054
50	CHENOPODIUM AMBROSIOIDES	0	2736